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PI Name:	Long, Roger M.D.		
Project Title:	Recovery of IGF-1 Signaling in Bone by Skeletal Reloading		
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PI Email:	Roger.Long@ucsf.edu	**Fax:**	FY 415-750-6929
PI Organization Type:	UNIVERSITY	**Phone:**	415-750-2089
Organization Name:	Northern California Institute for Research & Education		
PI Address 1:	4150 Clement St, 111N		
PI Address 2:			
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
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	POSTDOCTORAL FELLOWSHIP Prolonged spaceflight causes osteopenia due to decreased bone formation secondary to impaired osteoblast proliferation and increased osteoblast apoptosis. Hindlimb unloading by tail suspension, a model for skeletal unloading of spaceflight, causes osteoblast precursor and skeletal tissue resistance to the effects of insulin-like growth factor-1 (IGF-1). The nature of this resistance is characterized by decreased activation of the IGF-1 receptor and downstream signaling pathways. Osteoblast precursors isolated from unloaded bones demonstrate decreased expression of integrins, putative mechanosensors that bind extracellular matrix proteins, and in vitro treatment of normal osteoblasts with echistatin, an integrin antagonist, recreates the phenomena of unloading-induced IGF-1 resistance. Additionally, mechanical stimulation of human osteoblasts activates the IGF-1 receptor and augments the receptor response to IGF-1. These effects are abrogated by echistatin treatment. These findings suggest that integrin receptors have a role in mechanical		

Task Description:	regulation of IGF-1 receptor in osteoblasts. Our hypothesis is that interaction of integrin and IGF-1 receptor signaling cascades is required for IGF-1 activation of its receptor and intact IGF-1 signaling in osteoblasts. Mechanical loading stimulates the formation of an integrin/IGF-1 receptor complex, thus, enhancing IGF-1 signaling and enabling mechanically induced osteoblast proliferation and bone formation. To test the hypothesis, we proposed the following specific aims: 1. Determine the means by which skeletal reloading regulates skeletal response to IGF-1. Tail suspended rats treated with IGF-1 will be skeletally reloaded to preserve bone mass and osteoblast IGF-1 signaling. Integrity of IGF-1 signaling will be correlated with the interaction between integrins and the IGF-1 receptor. 2. Determine the mechanism for regulation of the osteoblast IGF-1 receptor by integrins in response to mechanical loading by pulsatile fluid flow. Formation of the integrin and IGF-1 receptor complex in human osteoblasts will be stimulated by fluid flow loading in order to determine the components within the complex and their function. At the completion of this project we established the critical regulatory role of integrins in IGF-1 signaling of osteoblasts. Mechanical stimulation of human osteosarcoma (HOS) cells activates the IGF-1R including the proliferative MAPK signal pathway and the anti-apoptotic Akt signal pathway in a ligand independent manner. The sensitivity of HOS cells and its IGF-1R to mechanical stimulation requires and is influenced by integrin receptor signaling. This is illustrated impaired activation of IGF-1R induced by echistatin and augmented activation of IGF-1R by stimulation of integrin receptors with exposure to their ligands. The results of siRNA knockdown experiments indicate that both integrin beta1 and beta3 are required for normal activation of the IGF-1R by treatment with either ligand or mechanical stimulation. The integrin regulation of IGF-1 signaling in osteoblasts allows osteoblasts to sense and respond to mechanical load, providing a link between mechanical loading, bone formation, and skeletal integrity. The feasibility of intermittent cyclic skeletal loading and IGF-1 infusion to prevent unloading-induced bone loss was evaluated. Tail suspended rats treated with intermittent short duration cyclic loading totaling only 42 minutes over 2 weeks had significantly blunted unloading induced trabecular bone loss. The mechanism of the enhanced response of the unloaded bones to cyclic loading is due to augmentation of bone formation rather than reduction in osteoclast activity and these responses were not affected by IGF-1 treatment. However, within the cortical bone compartment, the intermittent loading induced stimulation of periosteal bone formation was further augmented by treatment with IGF-1. These findings demonstrate the complex nature of the skeletal response to mechanical loading in that in cancellous bone, the threshold for mechanical stimulation induced bone formation in the unloaded state is lower that that required to prevent unloading induced osteoclast mediated bone loss and IGF-1 has a limited role in modulating the response of cancellous bone to mechanical loading but works synergistically with mechanical loading in cortical bone at the periosteal bone surface. The complexity of these differing skeletal responses to mechanical and pharmacologic treatments and the interactions between the integrin and IGF-1 signaling cascades in osteoblasts will need to be considered in the design of potential targets for countermeasure treatments to preserve the effects of mechanical loading and prevent the bone loss of spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research project outlined in my post-doctoral fellowship focuses on understanding the role of the hormone insulin-like growth factor 1 (IGF-1) in the skeletal response to skeletal unloading and loading. Previous work from our lab has demonstrated that skeletal unloading by tail suspension, a disuse osteoporosis model that not only mimics the weightlessness of spaceflight but the skeletal unloading associated with prolonged bedrest or neurological injury, causes a resistance to the anabolic effects of IGF-1 both in vivo and in vitro. A disruption of integrin signaling has been associated with the resistance of the IGF-1 receptor and the role of the interaction between these two signaling pathways will be explored. A membrane associated complex of the integrin and IGF-1 receptors would enable mechanically induced bone formation and osteoblast proliferation via enhanced IGF-1 signaling. Understanding the interaction between integrin receptors with their ability to sense mechanical load and the powerfully anabolic effects of IGF-1 is vital for efforts to maximize bone accrual and minimize bone loss associated with skeletal unloading. We propose to test the hypothesis that the skeletal response to mechanical load requires the formation of a complex including selected integrins and IGF-1R. If this hypothesis turns out to be valid, efforts to maintain the integrin signaling pathway will prove efficacious in preventing the skeletal loss of responsiveness to IGF-1 associated with skeletal unloading. We envision the use of intermittent skeletal loading, to maintain the integrity of the integrin and IGF-1 receptor interaction and thus the skeletal sensitivity to IGF-1, combined with IGF-1 therapy incorporated into hospital stays to prevent acute bone loss and rehabilitation programs to stimulate recovery of bone loss from acute injury or illness.
Task Progress:	Aim 1: During the second year of the post-doctoral fellowship multiple groups of rats were subjected to hindlimb unloading and concurrent episodic unilateral tibial axial compressive cyclic loading while receiving continuous IGF-1 or vehicle infusions. Six episodes of intermittent cyclic loading (40 compression events per episode) were administered concurrently with skeletal unloading by tail suspension over a two week period. Micro CT imaging and dynamic histomorphometry were utilized to study the effects of the various interventions. Application of intermittent cyclic loading during skeletal unloading by tail suspension blunts the profound bone loss associated with this disuse osteoporosis model. Without cyclic loading, the relative trabecular bone volume of the proximal tibiae decreased from 12% to 5% over the two-week skeletal unloading period. Only 42 minutes of intermittent cyclic compressive loading of the tibiae over a two week unloading period were able to diminish significantly this unloading induced bone loss. Histomorphometric analyses of this region revealed that skeletal unloading induced a profound reduction in osteoblast numbers and increase in osteoclast activity. Cyclic loading was able to stimulate recovery of only the osteoblast indices. IGF-1 infusion augmented the stimulatory effect of cyclic loading on osteoblast recovery but did not affect the recovery of relative trabecular bone volume in this short time period. Periosteal bone formation of the cortical bone was also profoundly reduced by unloading and only partially recovered with cyclic loading. However, IGF-1 infusion in addition to cyclic load normalized periosteal bone formation. These results demonstrate the possibility of maintaining skeletal integrity during the weightlessness of spaceflight with the use of only intermittent brief periods of mechanical stimulation supplemented with IGF-1. Aim 2:

	The osteoblast-like human osteosarcoma cells were studied to evaluate the role of integrins in the regulation of the IGF-1 receptor. Initial results demonstrate that stimulation of integrins with ligands that bind integrins beta1 or beta3 augments the phosphorylation of the IGF-1 receptor and activation of downstream signaling components in response to ligand and mechanical stimulation. Additionally, knockdown of integrins beta1 and/or beta3 by siRNA technology significantly impairs the activation of IGF-1 signaling suggesting that both these integrins play a significant regulatory role of IGF-1 signaling in osteoblasts.
Bibliography Type:	Description: (Last Updated: 07/29/2020)
Articles in Peer-reviewed Journals	Long RK, Halloran BP, Bikle DD. "Integrin regulation of the IGF-I receptor in bone, and the response to load." Clinical Reviews in Bone and Mineral Metabolism. 2007 Dec;5(4):222-33. http://dx.doi.org/10.1007/s12018-008-9009-3 , Dec-2007
Articles in Peer-reviewed Journals	Tahimic CG, Long RK, Kubota T, Sun MY, Elalieh H, Fong C, Menendez AT, Wang Y, Vilardaga JP, Bikle DD. "Regulation of ligand and shear stress-induced insulin-like growth factor 1 (IGF1) signaling by the integrin pathway." J Biol Chem. 2016 Apr 8;291(15):8140-9. https://doi.org/10.1074/jbc.M115.693598 ; PMID: 26865633; PMCID: PMC4825016 , Apr-2016
Awards	Long RK. "Alice L. Jee Young Investigator Award, 38th Sun Valley Workshop on Skeletal Biology, August 2008." Aug-2008